aws training and certification

Building Data Lakes on AWS - Build a Data Lake using AWS Lake Formation

End Lab | Open Console

Lab Information

Expires Mar 28 at 11:15 AM

45 minutes

Available languages

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Provisioning AWS services ...

Estimating the time remaining. Most labs take about 5 minutes, but it could take longer

Building Data Lakes on AWS - Build a Data Lake using AWS Lake Formation

SPL-CX-200-ANGBLA-1 – Version 1.0.0

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Note: Do not include any personal, identifying, or confidential information into the lab environment. Information entered may be visible to others.

Corrections, feedback, or other questions? Contact us at AWS Training and Certification.

Lab overview

You are a data engineer at AnyCompany, a cloud marketing organization. You are asked to run a pilot project to build a data lake using AWS Lake Formation.

In this lab, you work through the basic steps to register storage with Lake Formation and define a crawler to crawl the initial dataset. You then create an AWS Glue job to generate a version of the dataset in a partitioned Apache Parquet format. You also use Amazon Athena to validate the tables, their schema, and the ability to access the underlying data source through the data catalog.

Objectives

By the end of this lab, you will be able to do the following:

- Create a data lake and a database.
- Crawl data with AWS Glue to create a table.
- Query data using Athena.
- Transform data from .csv to Parquet format.

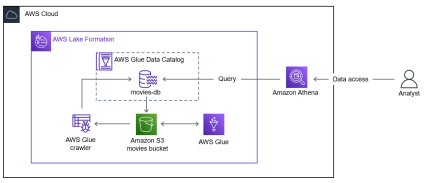
Icon key

Various icons are used throughout this lab to call attention to different types of instructions and notes. The following list explains the purpose for each icon:

- **Note:** A hint, tip, or important guidance.
- **Learn more:** Where to find more information.
- **Consider:** A moment to pause to consider how you might apply a concept in your own environment or to initiate a conversation about the topic at hand.
- **Hint:** A hint to a question or challenge.
- **Answer:** An answer to a question or challenge.

Environment overview

The following diagram shows the basic architecture of the lab environment:



In the preceding diagram, an Amazon Simple Storage Service (Amazon S3) bucket containing movie data is registered to Lake Formation to create a data lake. The data is loaded in AWS Glue. A crawler transforms the data from comma-separated values (.csv) to Parquet format and adds it to the database. The database is connected to the Amazon S3 bucket. An analyst is shown querying the Lake Formation database with Athena. Athena uses the AWS Glue Data Catalog and reads data from the database.

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.

 Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose **Open Console**.

 You are automatically signed in to the AWS Management Console in a new web browser tab.

 Warning: Do not change the **Region** unless instructed.

Common sign-in errors

Error: You must first sign out



If you see the message, **You must first log out before logging into a different AWS account:**

- Choose the **click here** link.
- Close your **Amazon Web Services Sign In** web browser tab and return to your initial lab page.
- Choose **Open Console** again.

Error: Choosing Start Lab has no effect

In some cases, certain pop-up or script blocker web browser extensions might prevent the **Start Lab** button from working as intended. If you experience an issue starting the lab:

- Add the lab domain name to your pop-up or script blocker's allow list or turn it off.
- Refresh the page and try again.

Task 1: Explore the lab environment

In this task, you review the account resources created before the lab started.

3. At the top of the AWS Management Console, in the search bar, search for and choose `S3`.
4. Choose the link for the bucket name that starts with **databucket**.

Note: The **data/** folder contains your dataset. The **results/** folder stores the results of your Athena queries. You will specify the **results/** folder as the query results location in Athena later in this lab.

5. Choose the **data/** folder. You will specify the **data/** folder as your designated storage location for your data lake later in this lab.
6. Choose the **movies_csv/** folder.

Note: There is one .csv file in the folder named **movies.csv**. It contains movie data from 1920-2016.

The file contains data similar to the following:

year	title	director_0	rating	genres_0	genres_1	rank	running_time_secs	actors_0	actors_1	actors_2	directors_1	directors_2
2013	Rush	Ron Howard	8.3	Action	Biography	2	7380	Daniel Brühl	Chris Hemsworth	Olivia Wilde		
2013	Prisoners	Denis Villeneuve	8.2	Crime	Drama	3	9180	Hugh Jackman	Jake Gyllenhaal	Viola Davis		
2013	The Hunger Games: Catching Fire	Francis Lawrence		Action	Adventure	4	8760	Jennifer Lawrence	Josh Hutcherson	Liam Hemsworth		
2013	Thor: The Dark World	Alan Taylor		Action	Adventure	5		Chris Hemsworth	Natalie Portman	Tom Hiddleston		
2013	This Is the End	Evan Goldberg	7.2	Comedy	Fantasy	6	6420	James Franco	Jonah Hill	Seth Rogen	Seth Rogen	

Congratulations! You successfully explored the lab environment.

Task 2: Set up Lake Formation

In this task, you register your Amazon S3 data storage and create a database.

Task 2.1: Register your Amazon S3 storage

Lake Formation manages access to designated storage locations in Amazon S3. Register the storage locations you want to be part of the data lake.

7. At the top of the AWS Management Console, in the search bar, search for and choose `AWS Lake Formation`.
8. In the **Welcome to Lake Formation** popup window, make sure **Add myself** is selected and then choose **Get started**.
9. In the left navigation pane, in the **Administration** section, choose **Data lake locations**.
10. Choose **Register location**.
11. On the **Register location** page, in the **Amazon S3 location** section:
 - For **Amazon S3 path**, copy and paste the **SourceDataLocation** value that is listed to the left of these instructions.
 - For **IAM role**, select **LakeFormationServiceRole**.

Note: The **LakeFormationService** role was created at the start of the lab and includes the **s3:PutObject**, **s3:GetObject**, **s3:DeleteObject**, and **s3:ListBucket** actions for your **DataBucket**.

12. Choose **Register location**.
13. In the left navigation pane, in the **Permissions** section, choose **Data locations**.
14. Choose **Grant**.
15. On the **Grant permissions** page, configure the following:
 - For **IAM users and roles**, select **AdminGlueServiceRole**.
 - For **Storage locations**, copy and paste the **SourceDataLocation** value that is listed to the left of these instructions.
16. Choose **Grant**.

Your Amazon S3 bucket **data/** folder is now registered as the storage location for your data lake.

Task 2.2: Create a database

Lake Formation uses the AWS Glue Data Catalog to store metadata about data lakes, data sources, transforms, and targets. Metadata about data sources and targets is in the form of databases and tables. Tables store information about the underlying data, including schema information, partition information, and data location. Databases are collections of tables.

Create a database in the AWS Glue Data Catalog.

17. In the left navigation pane, in the **Data catalog** section, choose **Databases**.
18. Choose **Create database**.
19. On the **Create database** page, in the **Database details** section:
 - For **name**, enter `movies-db`.
 - For **Location**, copy and paste the **SourceDataLocation** value that is listed to the left of these instructions.
20. Choose **Create database**.

You created a database in the AWS Glue Data Catalog using Lake Formation.

Congratulations! You successfully registered your Amazon S3 data storage and created a database.

Task 3: Crawl data with AWS Glue

In this task, you use an AWS Glue crawler to create a table for your movie data in your Lake Formation database.

Task 3.1: Create a crawler

A crawler connects to a data store and progresses through a prioritized list of classifiers to determine the schema for your data. It then creates metadata tables in your data catalog.

The crawler reads data at the source location and creates tables in the AWS Glue Data Catalog. A table is the metadata definition that represents your data, including its schema. The tables in the AWS Glue Data Catalog do not contain data. Instead, you use these tables as a source or target in an AWS Glue job definition.

21. At the top of the AWS Management Console, in the search bar, search for and choose `AWS Glue`.
22. In the left navigation pane, in the **Data catalog** section, choose **Tables**.
23. Choose **Add tables using crawler**.
24. For **Name**, enter `movies-table`.
25. Choose **Next**.
26. In the **Data sources** section, choose **Add a data source**.
27. For **S3 path**, copy and paste the **SourceDataLocation** value that is listed to the left of these instructions.

Note: With the default settings, the crawler will crawl all subfolders. The crawler will find the **movies.csv** file in the **movies_csv/** subfolder when it runs.

28. Choose **Add an S3 data source**.
29. Choose **Next**.
30. For **Existing IAM role**, choose **AdminGlueServiceRole**.
31. Choose **Next**.
32. For **Target database**, select **movies-db**.
33. Choose **Next**.
34. Choose **Create crawler**.

You created an AWS Glue crawler you can use to add data to your table.

Task 3.2: Run the crawler

Run your crawler to add data to your table.

35. Choose **Run crawler**.

The crawling task can take a few minutes to complete.

36. In the left navigation pane, in the **Data catalog** section, choose **Crawlers**.
37. Choose the refresh icon to get the current crawler status.

The **movies-table** status changes from **RUNNING** to **STOPPING** to **READY**.

38. Wait until the **State** is **Completed**.

You have run the crawler and added data to your table.

Task 3.3: View the table schema

Review the AWS Glue logs and view the new table schema created by the crawler.

39. Choose **View CloudWatch logs** to review the logs in Amazon CloudWatch, including logs related to data classification and table creation.

Consider: Take a moment to view the logs. Which event confirms the partitions for the table were created?

40. Return to the AWS Glue Console tab.
41. In the left navigation pane, in the **Data catalog** section, choose **Tables**.
42. Choose the refresh icon to get the current list of tables.

You see a **data** table.

Note: Since there is only one folder and file in the **data** folder, when the crawler runs it creates a table with the root folder name.


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